

How to Actually Program™

Part 1 — Computational Thinking

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Objectives

Welcome to CS 101!

- ▶ Meet your instructor and your TAs.
- ▶ Find the syllabus and getting started guide on the main web page.
- ▶ Open the lecture content on PrairieLearn and write some simple Python programs.
 - ▶ Print a greeting
 - ▶ Store some text into a variable
 - ▶ Store a number into a variable
- ▶ Start the post-class activity and get your first points.

Meet the Instructor!

- ▶ Name: Mattox Beckman
- ▶ History:
 - ▶ PhD, Fall 2003, University of Illinois at Urbana-Champaign
 - ▶ Lecturer 2003–2015 Illinois Institute of Technology
- ▶ Research Areas: CS Education, Programming Languages, Mathematical Foundations of Computer Science
- ▶ Professional Interests: Computer Science Education; Functional Programming; Mastery Learning
- ▶ Personal Interests: Irish Music; Philosophy; Evolution; Meditation; Kerbal Space Program; Home-brewing; Factorio ... and many many more ...

Meet the Staff!

- ▶ The TAs will be your main contact in this course. Let's meet some of them!

Meet You!

- ▶ Take 3 minutes to fill out this google form:
 - ▶ <https://forms.gle/cEQoxZpNebPSSybx6>

Find the Things!

- ▶ The course web site is at
`https://courses.engr.illinois.edu/cs101`
 - ▶ This link will redirect to a hosted location like `github` or `netlify`.
 - ▶ Things to see: the syllabus, the table of contents, the course schedule...
- ▶ Campuswire: (See syllabus)
- ▶ Prairielearn: `https://prairielearn.org`

Grading: Homework-like things

- ▶ Every lecture has attached to it a small assignment or lesson.
 - ▶ Part of it is done in class via a Jupyter Notebook.
 - ▶ Part of it is in a traditional Prairielearn question.
 - ▶ Collectively these are worth 5%.
- ▶ Once a week there is a homework set.
 - ▶ A bit larger, designed to help you program more interesting examples.
 - ▶ Collectively worth 35%
- ▶ You have one week. These can be extended though.

Grading: Labs

- ▶ There are labs sections
 - ▶ Two hours each week
 - ▶ **In person**, emphasizing group activities.
 - ▶ Most are in the CIF: self service has your details, and the course web page will be by the end of the week.
 - ▶ Collectively worth 25%.
- ▶ There is/will be a form to request a makeup if you miss your lab.

Grading: Quizzes and Final Exam

- ▶ There are six quizzes.
 - ▶ All in the Computer Based Testing Facility (CBTF).
 - ▶ Collectively worth 35%
- ▶ Email me if you are going to miss one.
- ▶ Final exam is **optional!**
 - ▶ Each quiz will have a “second chance” version.
 - ▶ You can take as few or many as you want.
 - ▶ if q is the quiz, s is the second chance, your new quiz score q' will be:

$$q' = \begin{cases} s & \text{if } s > q \\ \frac{q' + s}{2} & \text{otherwise} \end{cases}$$

Generative AI

- ▶ Your goal in this class is **not** to get points!
 - ▶ Unfortunately, that's what we can measure.
- ▶ You can use ChatGPT and friends, but:
 - ▶ You must cite it in your code if you use it.
 - ▶ You must be sure to check its work.
 - ▶ You must be sure to learn from its answers.

"If ChatGPT can do your job, ChatGPT will take your job."

Questions so far?

- ▶ Questions so far?
- ▶ Let's do some coding!